MODEL SUMMARY:

Machine Learning: EEG-Based Attention Classification

We trained a personalized-attention research prototype using the OpenNeuro EEG Meditation Study, containing recordings from 23 participants.

Feature extraction

- Four EEG electrodes: AF7, AF8, TP7 and TP8
- Five frequency bands: delta, theta, alpha, upper alpha and beta
- 20 log-transformed spectral-power features per observation
- EEG analyzed in the 30-second interval preceding questionnaire prompts

Model architecture

- Median imputation and feature standardization
- Supervised logistic regression with balanced class weights
- Participant-independent train/test split

Results

- 588 labeled EEG observations
- 20 participants used for training; 3 held out for testing
- 62.8% balanced accuracy
- 0.559 ROC-AUC
- 97.6% sensitivity for the higher-response class

Limitations and next steps

The model distinguishes provisional meditation questionnaire-response categories, which act as proxies for attention state. It is not yet a validated detector of focus loss during studying. Future improvements include verifying questionnaire labels, calibrating individual EEG baselines, and reducing false alerts through temporal smoothing and behavioral validation.